

GreenMind AI – Intelligent Environmental Sensor Recommendation System

GreenMind AI is an AI-powered decision support system developed for the Debrecen Green Sentinel environmental monitoring challenge. Rather than only visualizing existing measurements, the system recommends optimal locations for new air-quality monitoring stations by identifying areas with limited monitoring coverage and elevated environmental risk.

The application uses the official 30-day Green Sentinel air-quality dataset collected from 16 monitoring stations. During preprocessing, measurement files are cleaned by standardizing timestamps and station identifiers, converting invalid negative measurements to missing values, preserving incomplete observations, and transforming the raw long-format measurements into a processed wide dataset suitable for spatial analysis.

To estimate environmental conditions at locations without sensors, GreenMind AI applies **Inverse Distance Weighting (IDW)** interpolation using the five nearest monitoring stations. This estimates PM_{2.5}, PM₁₀, NO₂, O₃, wind speed, and historical variability for every candidate location across the city. Each location is then evaluated using four normalized criteria: **Coverage Gap (45%)**, **Pollution Risk (30%)**, **Historical Variability (15%)**, and **Low-Wind Risk (10%)**, producing a Priority Score between 0 and 100. The highest-ranked locations are selected while maintaining a minimum separation distance to avoid clustered recommendations.

Each recommendation includes an explainable AI summary, estimated environmental conditions, confidence score, nearest monitoring station, and supporting technical details. Confidence is calculated independently from priority by combining coverage certainty, pollution certainty, historical variability certainty, and wind certainty, enabling transparent interpretation of recommendation reliability.

A key innovation is the **dynamic simulation module**. Users can virtually deploy a new monitoring station, after which the frontend sends the updated network to the FastAPI backend. The backend recalculates interpolation, risk scores, confidence values, and the Top 5 recommended locations in real time, allowing planners to immediately observe how additional sensors improve monitoring coverage and reduce environmental blind spots.

The system follows a backend-driven architecture consisting of a React frontend, FastAPI backend, spatial recommendation engine, and processed Green Sentinel dataset. All recommendation logic is executed on the backend, ensuring consistency, scalability, and transparency while avoiding duplicated business logic in the frontend.

GreenMind AI transforms historical environmental measurements into actionable recommendations that help municipalities expand monitoring networks more effectively, prioritize investments, and make evidence-based environmental planning decisions.